

JEE MAIN REVISION MASTER: PROPERTIES OF MATTER SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 301 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Elasticity & Hooke's Law	137 PYQs	Solids
Very High	Surface Tension & Viscosity (Stokes' Law)	120 PYQs	Surface & Flow
Very High	Fluid Statics & Dynamics (Bernoulli & Torricelli)	48 PYQs	Fluids

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Elasticity & Hooke's Law	$Y = \frac{F/A}{\Delta L/L}$, $\Delta L = \frac{FL}{AY}$, $u = \frac{1}{2}\sigma\epsilon$, $\epsilon = \frac{\sigma^2}{2Y}$, $U = \frac{1}{2}F\Delta L$
Surface Tension & Viscosity (Stokes' Law)	$\Delta P_{\text{drop}} = \frac{2T}{R}$, $\Delta P_{\text{bubble}} = \frac{4T}{R}$, $h = \frac{2T\cos\theta}{r\rho g}$, $v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$
Fluid Statics & Dynamics (Bernoulli & Torricelli)	$A_1 v_1 = A_2 v_2$, $P + \frac{1}{2}\rho v^2 + \rho gh = \text{const}$, $v_{\text{efflux}} = \sqrt{2gh}$, $R = 2\sqrt{h(H-h)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Elasticity & Hooke's Law (Very High Yield — 137 PYQs)

Core Principle: Stress-strain relationships, Young's modulus, shear modulus, bulk modulus, and energy stored in stretched wires.

■ **Caution / JEE Pitfall:** Elongation due to self-weight is $\Delta L = (\rho g L^2 / 2Y) = (MgL / 2AY)$ (half of an equivalent end load Mg).

■ **Speed Trick:** When two wires are in series, equivalent Y is obtained by adding extensions: $\Delta L = \Delta L_1 + \Delta L_2$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is $4 \times 10^8 \text{ N/m}^2$ and i..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "Two wires as shown in the figure below, made of steel and have breaking stress of $12 \times 10^8 \text{ N/m}^2$. Area of cross-section of upper..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "A string A of length 0.314 m and Young's modulus $2 \times 10^{10} \text{ N/m}^2$ is connected to another string B of length and Young's modulu..." [Ans: B]

3.2 Surface Tension & Viscosity (Stokes' Law) (Very High Yield — 120 PYQs)

Core Principle: Surface energy, excess pressure in drops and bubbles, capillary rise formula, and terminal velocity of falling spheres.

■ **Caution / JEE Pitfall:** A soap bubble in air has TWO liquid-air interfaces, so excess pressure is $\Delta P = (4T / R)$, while a liquid drop has ONE ($\Delta P = (2T / R)$).

■ **Speed Trick:** Terminal velocity $v_t \propto r^2$. When n identical droplets coalesce: $R = n^{1/3}r$, and surface energy is released: $\Delta U = 4\pi T r^2(n - n^{2/3})$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (Numerical): "The surface tension of a soap solution is $3.5 \times 10^{-2} \text{ N/m}$. The work required to increase the radius of a soap bubble from 1 cm to 2..." [Ans: 264]
- [2026 Morning Shift] Q#1 (Numerical): "Sixty four rain drops of radius 1 mm each falling down with a terminal velocity of 10 cm/s coalesce to form a bigger drop. The terminal vel..." [Ans: 160]
- [2026 Evening Shift] Q#2 (MCQ): "A spherical liquid drop of radius R acquires the terminal velocity $v_t(1)$ when falls through a gas of viscosity η . Now the drop is ..." [Ans: D]

3.3 Fluid Statics & Dynamics (Bernoulli & Torricelli) (Very High Yield — 48 PYQs)

Core Principle: Continuity equation, Bernoulli's energy conservation principle, Torricelli's efflux theorem, and hydraulic lift.

■ **Caution / JEE Pitfall:** Speed of efflux is $\sqrt{2gh}$ where h is depth from the free liquid surface, NOT height from the ground!

■ **Speed Trick:** Maximum horizontal range of water jet occurs when hole is at mid-depth $h = H/2$, with $R_{\text{(max)}} = H$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A liquid of density 600 kg/m^3 flowing steadily in a tube of varying cross-section. The cross-section at a point A is 1.0 cm^2 and t..." [Ans: D]
- [2026 Evening Shift] Q#2 (MCQ): "A cubical block of density $\rho_b = 600 \text{ kg/m}^3$ floats in a liquid of density $\rho_e = 900 \text{ kg/m}^3$. If the height of block is $H = 8.0 \text{ cm}$..." [Ans: D]
- [2026 Morning Shift] Q#2 (MCQ): "Water flows through a horizontal tube as shown in the figure. The difference in height between the water columns in vertical tubes is 5 cm a..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A liquid of density 600kg/m^3 flowing steadily in a tube of varying cross-section. The cross-section at a point A is 1.0cm^2 and that at B is 20mm^2 . Both the points A and B are in same horizontal plane, the speed of the liquid at A is 10cm/s . The difference in pressures at A and B points is ____ Pa.

- (A) 18
- (B) 144
- (C) 36
- (D) 72 [Correct]

Correct Answer: D

Solution Analysis: The density of liquid is $\rho=600\text{kg/m}^3$. Area of cross-section at point A is $A_A=1.0\text{cm}^2=1.0 \times 10^{-4}\text{m}^2$. Area of cross-section at point B is $A_B=20.0\text{mm}^2=20.0 \times 10^{-6}\text{m}^2=0.2 \times 10^{-4}\text{m}^2$. The velocity at A is $v_A=10\text{cm/s}=0.1\text{m/s}$. For an incompressible liquid in steady flow, the mass flow rate must remain constant. Using equation of continuity, $A_A v_A = A_B v_B$. Substituting the values : $(1.0 \times 10^{-4}) \times (0.1) = (0.2 \times 10^{-4}) \times v_B \Rightarrow v_B = (1.0 \times 0.1 / 0.2) = (0.1 / 0.2) = 0.5\text{m/s}$ According to

Benchmark Problem 2: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A lift of mass 1600kg is supported by thick iron wire. If the maximum stress which the wire can withstand is $4 \times 10^8\text{N/m}^2$ and its radius is 4mm , then maximum acceleration the lift can take is ____ m/s^2 . (take $g=10\text{m/s}^2$ and $\pi=3.14$)

- (A) 2.56 [Correct]
- (B) 3.89
- (C) 4.32
- (D) 5.16

Correct Answer: A

Solution Analysis: When a lift of mass M accelerates upward with acceleration a , the tension (T) in the supporting cable must overcome both gravity and provide the upward acceleration. Using Newton's Second Law ($F_{\text{net}}=Ma$): $T-Mg=Ma \Rightarrow T=M(g+a)$ The stress (σ) is defined as the restoring force (tension) per unit cross-sectional area (A): $\sigma=(T/A)=(M(g+a)/A)$ For a wire with a circular cross-section of radius r , the area of cross-section is $A=\pi r^2$. $\sigma_{\text{(max)}}=\frac{M(g+a_{\text{(max)}})\pi r^2}{\pi r^2}$ The mass of lift is $M=1600\text{kg}$ The radius of cable is $r=4\text{mm}=4 \times 10^{-3}\text{m}$

Benchmark Problem 3: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

The surface tension of a soap solution is $3.5 \times 10^{-2}\text{N/m}$. The work required to increase the radius of a soap bubble from 1cm to 2cm is $\alpha \times 10^{-6}\text{J}$. The value of α is ____ . ($\pi=22/7$)

Correct Answer: 264

Solution Analysis: A soap bubble has two free surfaces (an inner surface and an outer surface). The surface energy (U) of a bubble with radius r and surface tension T is: $U=T \times (\text{Total Surface Area}) \Rightarrow U=T \times (2 \times 4\pi r^2)=8\pi r^2 T$ The work done (W) in increasing the radius from r_1 to r_2 is equal to the increase in surface energy: $W=\Delta U=8\pi (r_2^2 - r_1^2) T$ The surface tension of soap solution is given as $T=3.5 \times 10^{-2}\text{N/m}$ The initial radius of soap bubble is $r_1=1\text{cm}=10^{-2}\text{m}$ The final radius of soap bubble is $r_2=2\text{cm}=2 \times 10^{-2}\text{m}$